

Assignment 1

CSC115

Due Date: September 6, 11:55. **Points: 100.**

Single PDF file named: A1_*.pdf where * is your last name.

A correct answer with no work shown is worth no points, but an incorrect or partial answer with some work shown MAY be worth partial credit.

1- Convert the following binary numbers to decimal. Show your work (15 points).

a. 1010_2

1a.) $1010 = 2^1(2) + 2^3(8) = 10$

b. 110110_2

1b.) $110110 = 2^5(32) + 2^4(16) + 2^2(4) + 2^1(2) = 54$

c. 11110000_2

1c.) $11110000 = 2^7(128) + 2^6(64) + 2^5(32) + 2^4(16) = 240$

d. 000100010100111_2

1d.) $000100010100111 = 2^{11}(2048) + 2^7(128) + 2^5(32) + 2^2(4) + 2^1(2) + 2^0(1) = 2215$

2- Convert the following binary numbers to octal. Show your work. (15 points).

a. 111110110_2

*Add zeros to the left of the last digit if there aren't enough digits to make a set of three.

2a.) $111110110 =$
 $(111) = 7$ $(110) = 6$ $(110) = 6$
 $= 766$

b. 1000001_2

2b.) $1000001 =$
 $(001) = 4$ $(000) = 0$ $(001) = 4$
 $= 404$

c. 10000010_2

2c.) $10000010 =$
 $(010) = 2$ $(000) = 0$ $(010) = 2$
 $= 202$

d. 1100010_2

2d.) $1100010 = (001) = 1$ $(100) = 4$ $(010) = 2$ $= 142$

3- Convert the following binary numbers to hexadecimal. Show your work. (15 points).

a. 10101001_2

*Add zeros to the left of the last digit if there aren't enough digits to make a set of four:

3a.) $10101001 =$
 $(1010) = A$ $(1001) = 9$
 $= A9$

b. 11100111_2

3b.) $11100111 =$
 $(1110) = E$ $(0111) = 7$
 $= E7$

c. 01101110_2

3c.) $01101110 =$
 $(0110) = 6$ $(1110) = E$
 $= 6E$

d. 01111111_2

3d.) $01111111 =$
 $(0111) = 7$ $(1111) = F$
 $= 7F$

$$\begin{array}{l}
 4a.) 42 = (42 - 32) = 10 \text{ (1)} \\
 0 \\
 (10 - 8) = 2 \text{ (1)} \\
 0 \\
 (2 - 2) = 0 \text{ (1)} \\
 0 \\
 = 101010
 \end{array}$$

$$\begin{array}{l}
 4b.) 63 = (63 - 32) = 31 \text{ (1)} \\
 (31-16) = 15 \text{ (1)} \\
 (15-8) = 7 \text{ (1)} \\
 (7-4) = 3 \text{ (1)} \\
 (3-2) = 1 \text{ (1)} \\
 (1-1) = 0 \text{ (1)} \\
 = 111111
 \end{array}$$

$$\begin{array}{l}
 4c.) 229 = (229-128) = 101 \text{ (1)} \\
 (101-64) = 37 \text{ (1)} \\
 (37-32) = 5 \text{ (1)} \\
 0 \\
 0 \\
 (5-4) = 1 \text{ (1)} \\
 0 \\
 (1-1) = 0 \text{ (1)} \\
 = 11100101
 \end{array}$$

4- Convert the following decimal numbers to binary numbers. Show your work. **(15 points).**

- a. 42_{10}
- b. 63_{10}
- c. 229_{10}
- d. 845_{10}

$$\begin{array}{l}
 4d.) 845 = (845-512) = 333 \text{ (1)} \\
 (333-256) = 77 \text{ (1)} \\
 0 \\
 (77-64) = 13 \text{ (1)} \\
 0 \\
 0 \\
 (16-13) = 3 \text{ (1)} \\
 (3-2) = 1 \text{ (1)} \\
 0 \\
 (1-1) = 0 \text{ (1)} \\
 = 1101001101
 \end{array}$$

5- Convert the following decimal numbers to octal. Show your work. **(15 points).**

$$\begin{array}{l}
 5a.) 777 = (8/777) = 97R1 \\
 (8/97) = 12R1 \\
 (8/12) = 1R4 \\
 (8/1) = 0R1 \\
 = 1411
 \end{array}$$

- a. 777
- b. 605
- c. 443
- d. 1

$$\begin{array}{l}
 5c.) 443 = (8/443) = 55R3 \\
 (8/55) = 6R7 \\
 (8/6) = 0R6 \\
 = 673
 \end{array}$$

$$\begin{array}{l}
 5d.) 1 = (8/1) = 0R1 \\
 = 1
 \end{array}$$

$$\begin{array}{l}
 5b.) 605 = (8/605) = 75R5 \\
 (8/75) = 9R3 \\
 (8/9) = 1R1 \\
 (1/8) = 0R1 \\
 = 1135
 \end{array}$$

6- Convert the following decimal numbers to hexadecimal. Show your work. **(15 points).**

$$\begin{array}{l}
 6a.) 777 = (16/777) = 48R9 \\
 (16/48) = 3R0 \\
 (16/3) = 0R3 \\
 = 309
 \end{array}$$

- a. 777
- b. 605
- c. 443
- d. 1

$$\begin{array}{l}
 6b.) 605 = (16/605) = 37R13 \\
 (16/37) = 2R5 \\
 (16/2) = 0R2 \\
 = 25D
 \end{array}$$

$$\begin{array}{l}
 6d.) 1 = (16/1) = 0R1 \\
 = 1
 \end{array}$$

$$\begin{array}{l}
 6c.) 443 = (16/443) = 27R11 \\
 (16/27) = 1R11 \\
 (16/1) = 0R1 \\
 = 1BB
 \end{array}$$

7- How many different numbers can be represented with 16 bits? **(10 points).**

16 bits can hold up to 2^{16} different values, which is 65,536.
Range = -32,768 to +32,767.